Phase 5 Contribution and AI sheet.
Phase 5 due 3/15/2026

The total number of files is 10 + updated assembler, so 11 total .v files.

Pipeline Registers

- `IF_ID.v` — holds PC and instruction between Fetch and Decode
- `ID_EX.v` — holds control signals, register values, immediates between Decode and Execute
- `EX_MEM.v` — holds ALU result, write data, control signals between Execute and Memory
- `MEM_WB.v` — holds memory/ALU result and control signals between Memory and Writeback

Hazard Handling

- `FORWARDING_UNIT.v` — detects RAW hazards and generates ForwardA/ForwardB select signals
- `HAZARD_DETECTION.v` — detects load-to-use hazards and generates stall/flush signals

New Muxes

- `MUX_3_1.v` — you'll need 3-to-1 muxes for the forwarding paths into the ALU (ForwardA and ForwardB selection)

Existing files you'll likely need to significantly modify:

- `RISCV_TOP.v` — rewire everything to connect pipeline stages
- `CONTROL.v` — control signals now need to flow through pipeline registers
- `REGISTER.v` — may need adjustments for stalling

That's about 7 new files. Want to start with the pipeline registers since everything else depends on them?

Amanda - MUX_3_1.v file, IF_ID.v file, Contribution/AI document

Sophia - REGISTER.v file, HAZARD_DETECTION.v file, FORWARDING_UNIT.v file

Evan - CONTROL.v file, ID_EX.v file, Testing and submission

Banmeet - RISCV_TOP.v file, MEM_WB.v file, EX_MEM.v file

AI usage report.
Due 3/15/26
Phase 5

For Phase 5, we wanted to double check what files we needed in order to fully implement the Multi Cycle Processor. We asked Claude AI what files we would need in this Phase to complete the Pipeline as shown below:

Pipeline Registers

- `IF_ID.v` — holds PC and instruction between Fetch and Decode
- `ID_EX.v` — holds control signals, register values, immediates between Decode and Execute
- `EX_MEM.v` — holds ALU result, write data, control signals between Execute and Memory
- `MEM_WB.v` — holds memory/ALU result and control signals between Memory and Writeback

Hazard Handling

- `FORWARDING_UNIT.v` — detects RAW hazards and generates ForwardA/ForwardB select signals
- `HAZARD_DETECTION.v` — detects load-to-use hazards and generates stall/flush signals

New Muxes

- `MUX_3_1.v` — you'll need 3-to-1 muxes for the forwarding paths into the ALU (ForwardA and ForwardB selection)

Existing files you'll likely need to significantly modify:

- `RISCV_TOP.v` — rewire everything to connect pipeline stages
- `CONTROL.v` — control signals now need to flow through pipeline registers
- `REGISTER.v` — may need adjustments for stalling

That's about 7 new files. Want to start with the pipeline registers since everything else depends on them?

We also wanted skeleton files to have a guide in order to head in the right direction with our Multi-Cycle Processor, so we asked Claude to give us sample skeleton files that may go into the processor. With the skeleton files Claude produced we went in and fixed the logic and tailored them to the requirements of our project.
We also use AI to double check for syntax errors and if we need help with testing before submission.